

DSA Day 17: Recursion on strings & lists

Book: Data Structures Through C in Depth - Ch5 §5.5 · pp 163-166 | Code in Python

Overview

Recursion naturally processes strings and linked lists by handling the first element and recursing on the rest.

Key concepts to master

- Unwinding order
- Implicit stack
- Recursive reverse / print

Python implementation

```
def rev(s):  
    return s if len(s)<=1 else rev(s[1:])+s[0]  
def print_rev(node):  
    if node: print_rev(node.next); print(node.data)
```

Complexity

O(n) time, O(n) stack.

Common pitfalls

- Building huge intermediate strings
- Stack overflow on long lists

Practice

- Reverse 'hello'
- Print a list backwards

